

Hybrid Question Answering over Graph and Textual Evidence

M2 Internship — LaHDAK

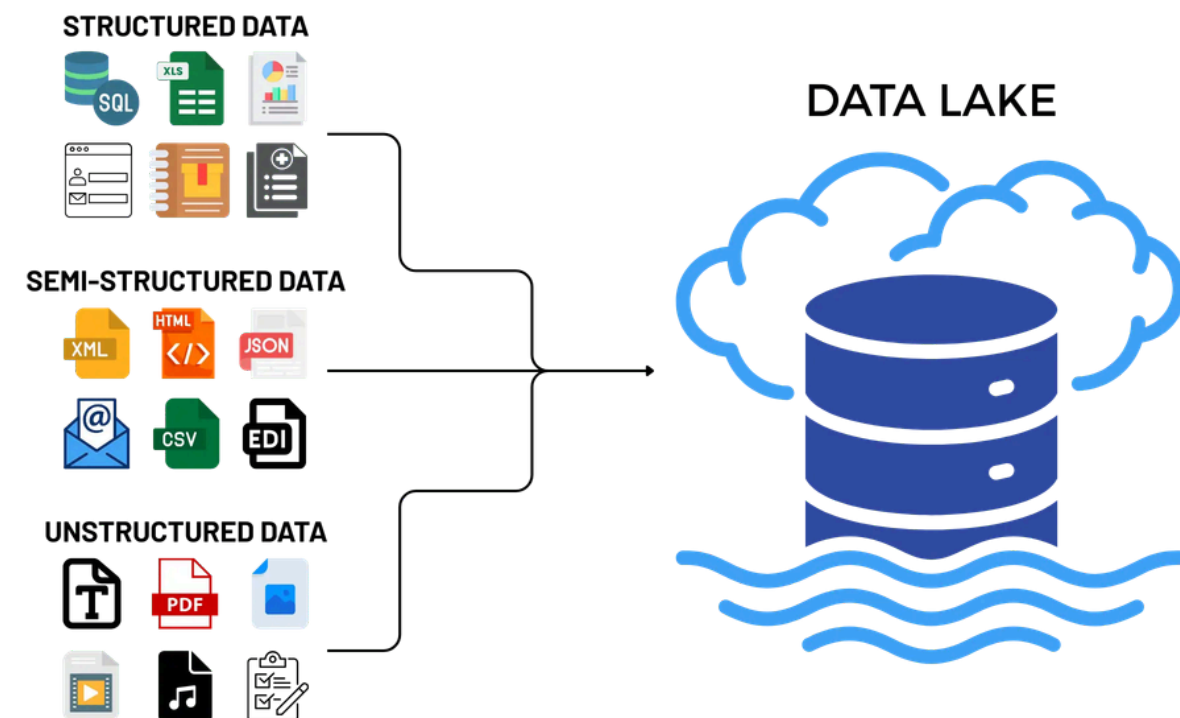
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Why is heterogeneous data difficult to exploit?

From heterogeneous data lakes to our scope :



Journalist : “Can I ask one question across all these sources?”

What the user needs

Answer :

- Evidence
- Provenance

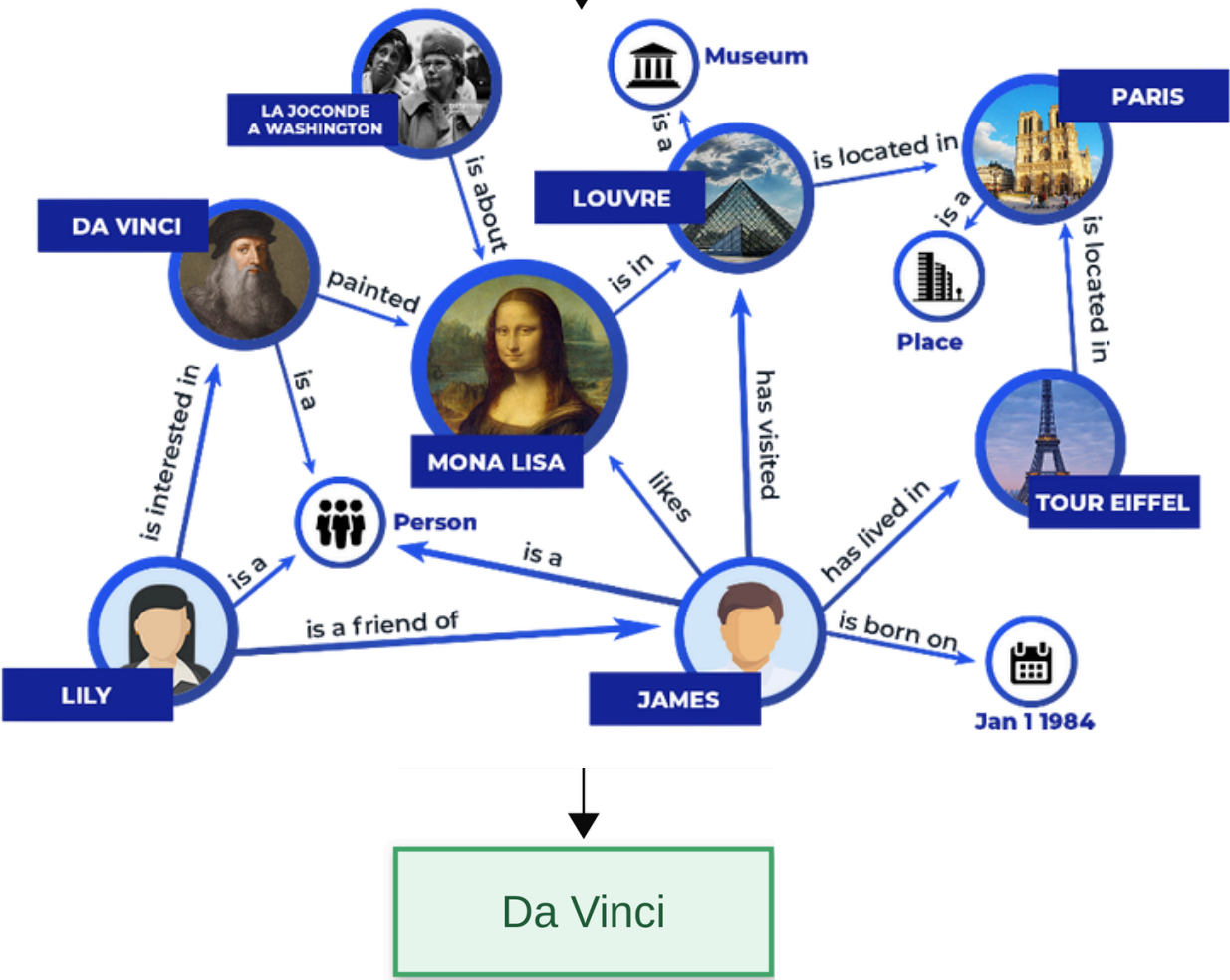
➡ *This work focuses on two complementary evidence sources: Knowledge Graphs + Text*

Two complementary ways of answering a question

Knowledge Graph Question Answering (KGQA)

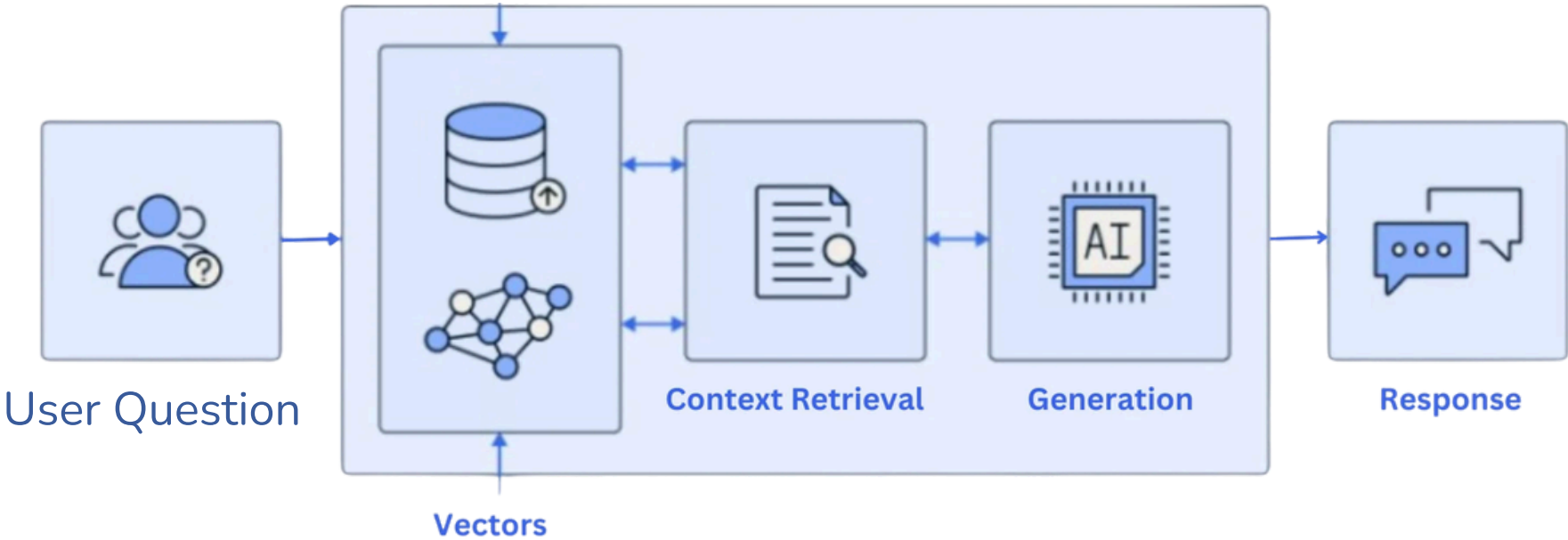
User Question :

Who painted the Mona Lisa?



Da Vinci

RAG over Text



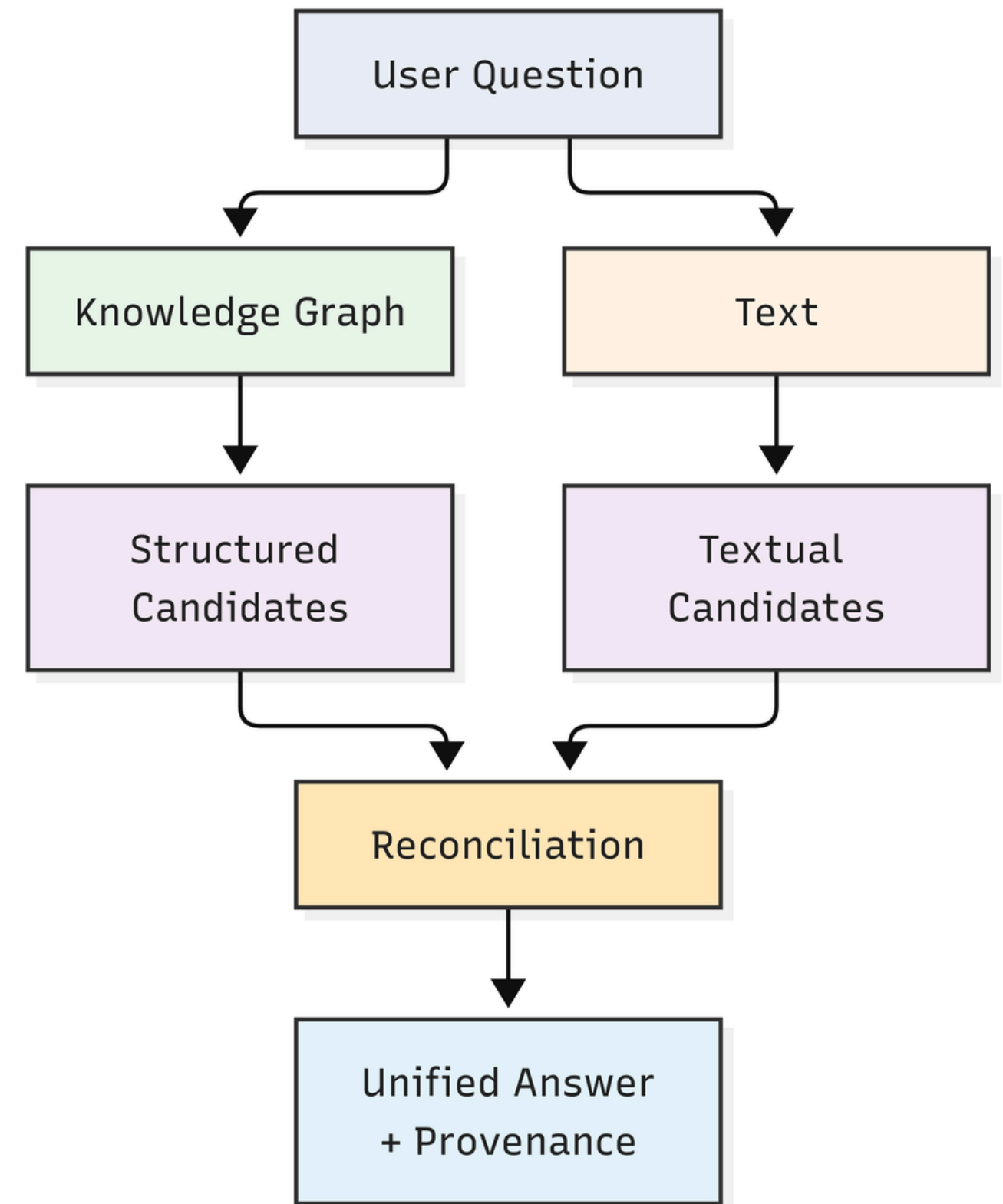
Research Problem

When a question is evaluated :

➡ both components produce different candidate results in different formats

Core scientific problem:

➡ How can independently produced graph and textual evidence be compared and reconciled without losing provenance?



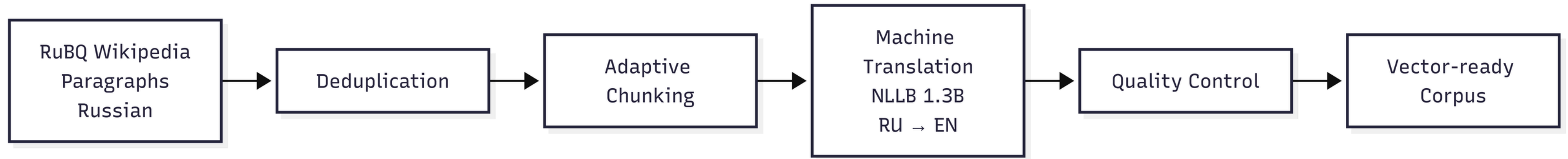
Choosing the right benchmark

To study both evidence sources , we need graph and textual data for the same questions.

Benchmark	NL Questions	Structured Query	Answers	Aligned Text Evidence
WebQSP	✓	✓	✓	✗
ComplexWebQuestions	✓	✓	✓	✗
RuBQ 2.0 : a Russian dataset for KBQA over Wikidata	✓	✓	✓	✓

⚠ Building a usable textual evidence channel required translating the Russian paragraphs in English .

Building the English textual corpus



SELECTED FOR THE PIPELINE : **NLLB 1.3B**

Best local compromise :

- ✓ Strong translation quality
- ✓ Fully local & reproducible
- ✓ Scales to the full corpus

<i>Model</i>	<i>COMETKiwi</i>
DeepL	0.8511
NLLB 1.3B	0.8378
NLLB 600M	0.8268
Helsinki	0.8072

⚠ Translation can introduce lexical and semantic shifts, potentially biasing retrieval results .

Retrieving Relevant Passages : Does translation preserve retrieval quality?

Same retriever, three controlled conditions

multilingual-E5-base : embedding model

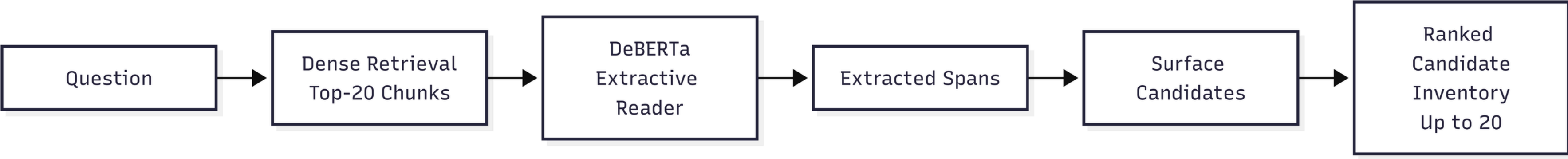
FAISS : vector similarity search index

Condition	Hit@10	MRR
RU → RU	91.79%	0.737
EN-NLLB → EN-NLLB	89.30%	0.689
EN-ref → EN-NLLB	88.81%	0.700

Insight : Translation introduces a small retrieval loss but the English channel remains highly usable !

 But retrieval is only the first step : can we produce the right answer?

Extracting the answer from the relevant passages



Chunk:

"The capital of Germany is the city state of Berlin..."

└───┘
span

surface = "Berlin"

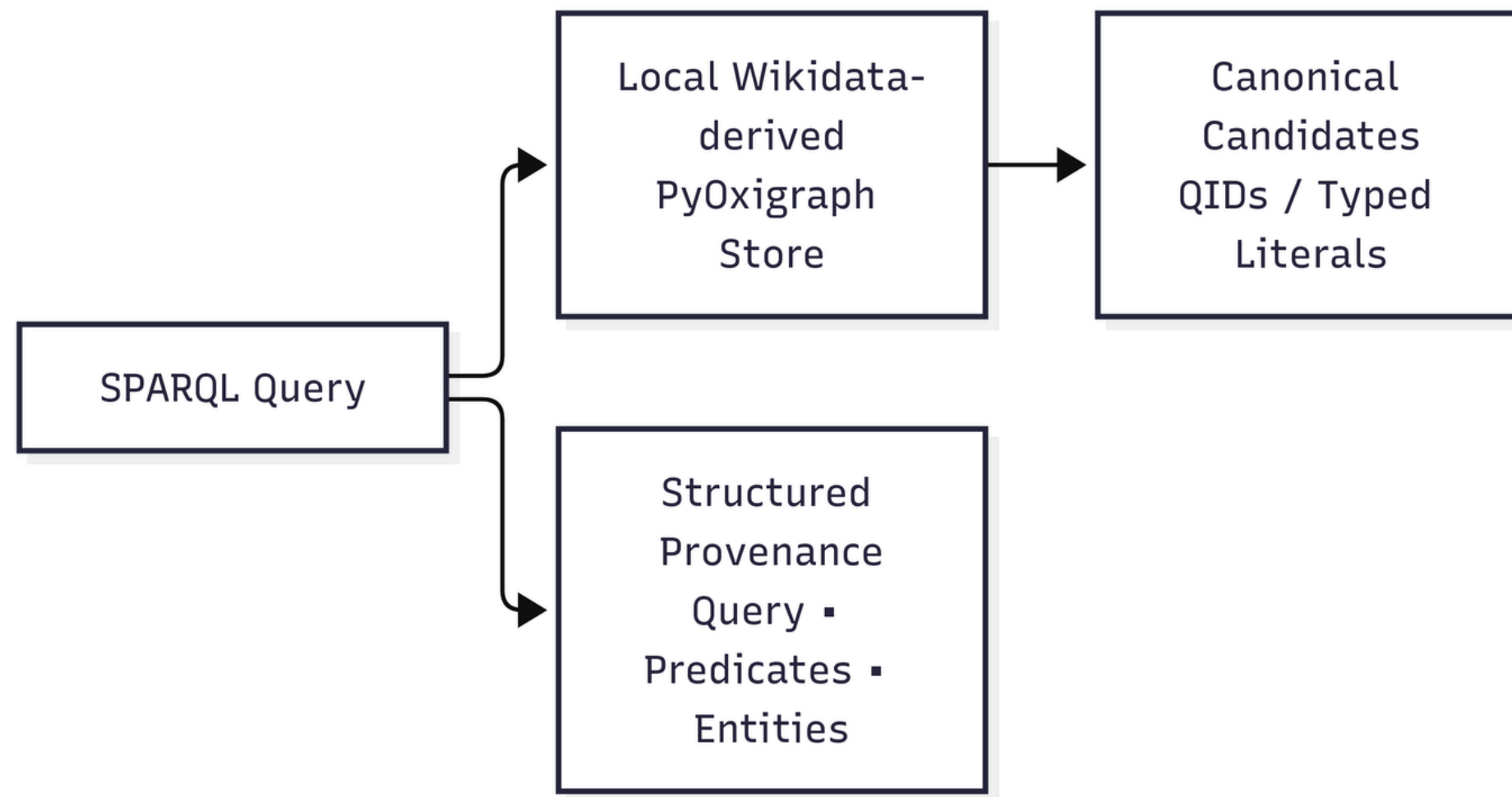
Evidence:

chunk + supporting sentence + offsets + scores : **important for provenance purposes**

Metric	Score
Exact Match @1	48.0%
Token F1	58.6%
Hit@20	64.4%

Building the structured evidence channel

⚠ Can the SPARQL queries provided by the dataset reliably reproduce the expected answers on our local graph ?



Controlled setting :

- SPARQL query provided by the dataset is being used .
- **96.75%** of executable gold queries reproduce all expected answers in the local store

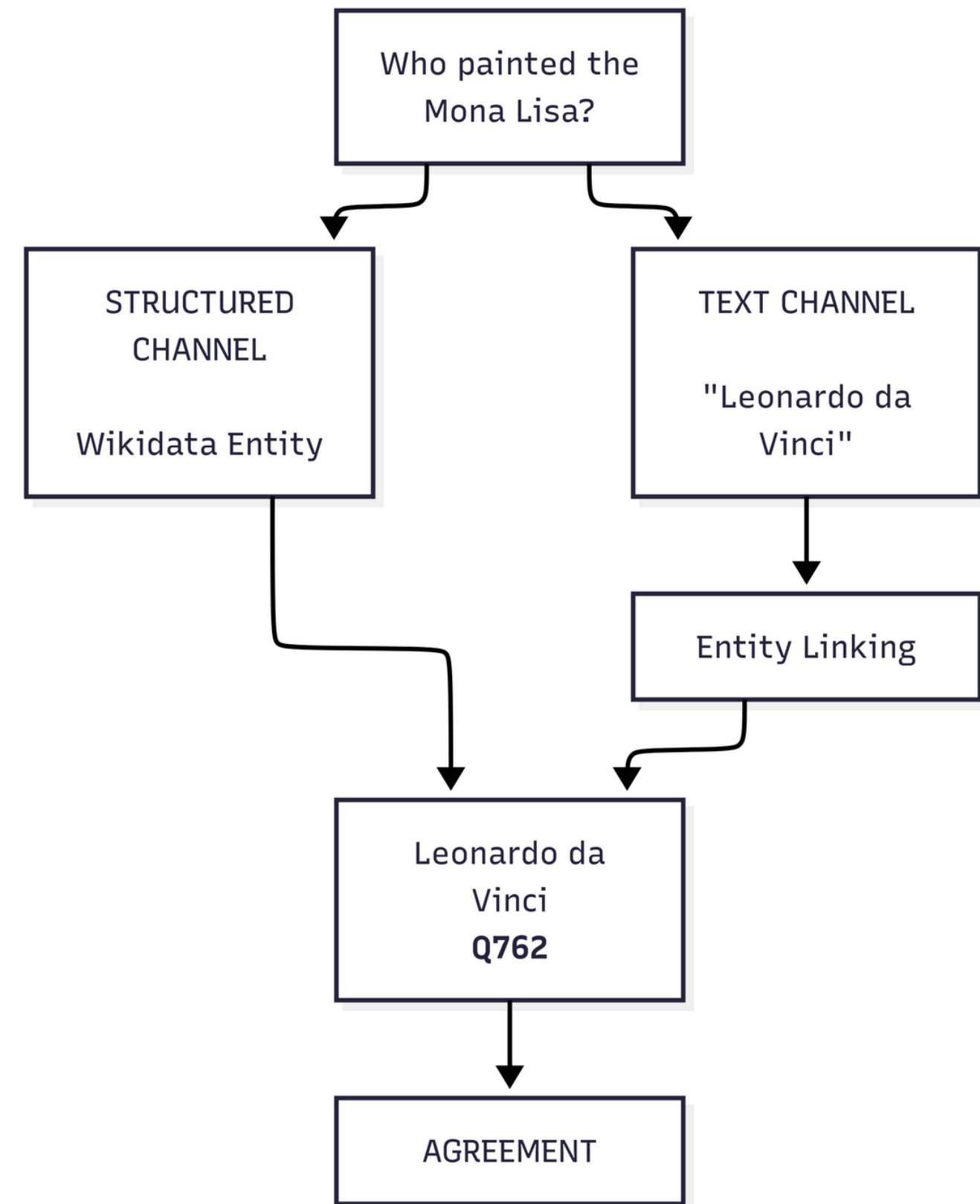
Purpose:

➡ isolate graph execution from query-generation errors

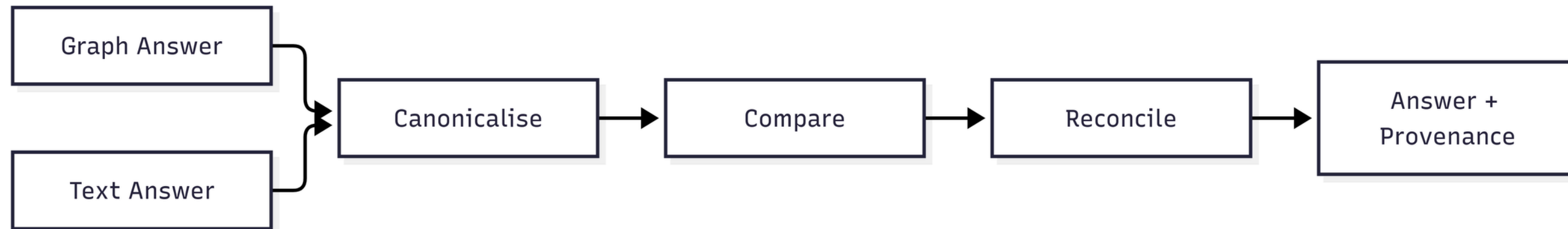
Now both channels exist: how can we reconcile their results?

Towards reconciliation: first make the outputs comparable

- Normalize both channels into comparable representations.
- Study agreement, disagreement and provenance.
- Produce final answer with sources .



From two channels to hybrid reconciliation



Next research questions

- How can graph and text answers be mapped to a common representation?
- How should agreement, disagreement and missing evidence be handled?
- How can the final answer preserve its supporting evidence and provenance?